EXPERIMENT 14

Pro code extension with Catalyst

Aim

Extend the library app with a small authenticated Catalyst service and compare its value with a simpler Creator custom function.

Procedure

1. Select a bounded extension: validate an ISBN-13 and return its canonical digits and checksum result. This demonstrates a testable external function without delegating loan authorization or exposing member data.

2. Create an authorized Catalyst development project and an Advanced I/O function using a supported Node.js runtime. Use the generated project skeleton and dependency file. Add a POST route /validate-isbn with JSON input containing isbn.

3. Configure authentication through supported Catalyst security rules or API Gateway. Verify the exact mechanism in the project. Never leave the function anonymously callable merely because the endpoint is difficult to guess.

4. Implement the pure validation function below, then call it from the generated HTTP handler. Accept a string, remove spaces and hyphens, enforce 13 digits and calculate the check digit. No database is needed.

```
// JavaScript pure function used by the HTTP handler.
function validateISBN13(value) {
 if (typeof value !== "string") {
    return { valid: false, reason: "isbn must be text" };
  }
 const isbn = value.replace(/[ -]/g, "");
if (!/^\d{13}$/.test(isbn)) {
    return { valid: false, reason: "Use 13 digits" };
  }
 let sum = 0;
 for (let i = 0; i < 12; i++) {
    sum += Number(isbn[i]) * (i % 2 === 0 ? 1 : 3);
  }
 const check = (10 - (sum % 10)) % 10;
return {
    isbn,
    valid: check === Number(isbn[12])
 };
}
```

This checks an ISBN-13 checksum, not whether a book exists or whether an ISBN was assigned by an agency. Label the UI accordingly. The HTTP wrapper must also validate content type, request size and authorization, and return a consistent JSON response.

Interface contract

Request: POST /validate-isbn with {"isbn":"9780306406157"}. A valid request returns HTTP 200 and {"isbn":"9780306406157","valid":true}. Malformed input returns a documented 400 response; unauthenticated or unauthorized access returns 401 or 403 according to the gateway configuration.

Implementation and verification

Procedure continued

1. Create a compatible authenticated connection or approved server-side authentication flow in Creator. Store credentials securely. Confirm that the chosen connection's token type and scopes are actually accepted by the Catalyst endpoint.

2. Use a staff-only Validate ISBN action for Books. Supply only the ISBN and display the returned checksum result. Do not call the service on every keystroke, and do not let an external timeout block unrelated circulation.

3. Replace the URL and connection below with the deployed development values. The endpoint is a placeholder contract, not a ready-to-use live address. Add transport, JSON-shape and record-level error handling.

```
// Deluge function body: isbnText is a string argument.
body = Map();
body.put("isbn", isbnText);
reply = invokeurl
[


    url :"https://<verified-catalyst-host>/server/"
         + "<function-name>/validate-isbn"
    type :POST
    parameters:body.toString()
    headers:{"Content-Type":"application/json"}
    connection:"library_catalyst"
];
// Check response shape before displaying isbn and valid.
// Handle transport errors separately from valid == false.
```

Test input or conditio	Expected behavior
9780306406157	Checksum true.
9780306406158	Checksum false.
978-0-306-40615-7	Normalize; checksum true.
Null, number or fewer than 13 digits	Reject under the documented input contract.
Missing or invalid credentials	Denied; function not publicly usable.
Service timeout or malformed JSON	Readable temporary error; no book or loan corruption.

Tradeoff and deployment

This checksum can also run in a Creator custom function without a network hop. Catalyst is a learning exercise here; specialized libraries or heavier workloads may justify it in practice. Compare latency, security and maintenance effort.

Deploy the tested function using separate development and production configuration. Version the interface, restrict allowed methods, cap request sizes and monitor failures. Calling a Catalyst function does not automatically make Creator writes atomic; the allocation problem from Experiment 13 requires a different authoritative design.

Illustrative process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

The checksum service is deliberately small so its input and output can be tested clearly. An ISBN checksum does not confirm the title, publisher or ownership of a book. Prefer a local custom function when a separate service adds no operational benefit.

Result

Expected result and verification criteria. E14 contains the function, interface contract, authenticated Creator call and test results. Valid and invalid checksums are distinguished from service failures, and the student can justify when an external service is worth using.